



RENDERED.AI

The Platform for Accelerated Synthetic Data Generation

USGIF
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SYMPOSIUM 2025



Validating Satellite Data Collection Before Deployment

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The AI Data Gap in Satellite Development

- The “Catch-22” Challenge: Relying on orbital data to build a reliable model that is not yet collecting that data.
- Consequences for satellite reliability & value to end-users (e.g. mission failure, sub-optimal performance, additional development time & costs).
- Betting billions on unproven satellite technologies, typically \$50-\$500 million in development & design costs alone.
- Ongoing pressure to train & adjust models on the ground.
- Train AI tasks on Earth with high confidence, deploy, then collect real data to support more complex tasks later.



A Formula for Success

The Effective Use of Synthetic Data:

- Automated workflows to rapidly generate test data that accurately mimics in-orbit data collection
- Rapidly scalable training datasets with increased diversity
- Physics-based engineering approach
- Direct integration into AI pipelines

Results:

- Reduced time to validate data collection - 6-9 months less on average
- Successful demonstration of data collection to stakeholders before deployment
- A significant improvement in AI performance well before launch
- Lower resource, expense, & personnel training requirements



Synthetic Data in Action: Planet Labs



USE CASE: Testing software pipelines for the Tanager hyperspectral satellite constellation with simulated data to effectively detect & track methane & CO2 super-emitters at a level of granularity needed to support direct mitigation actions prior to constellation launch.

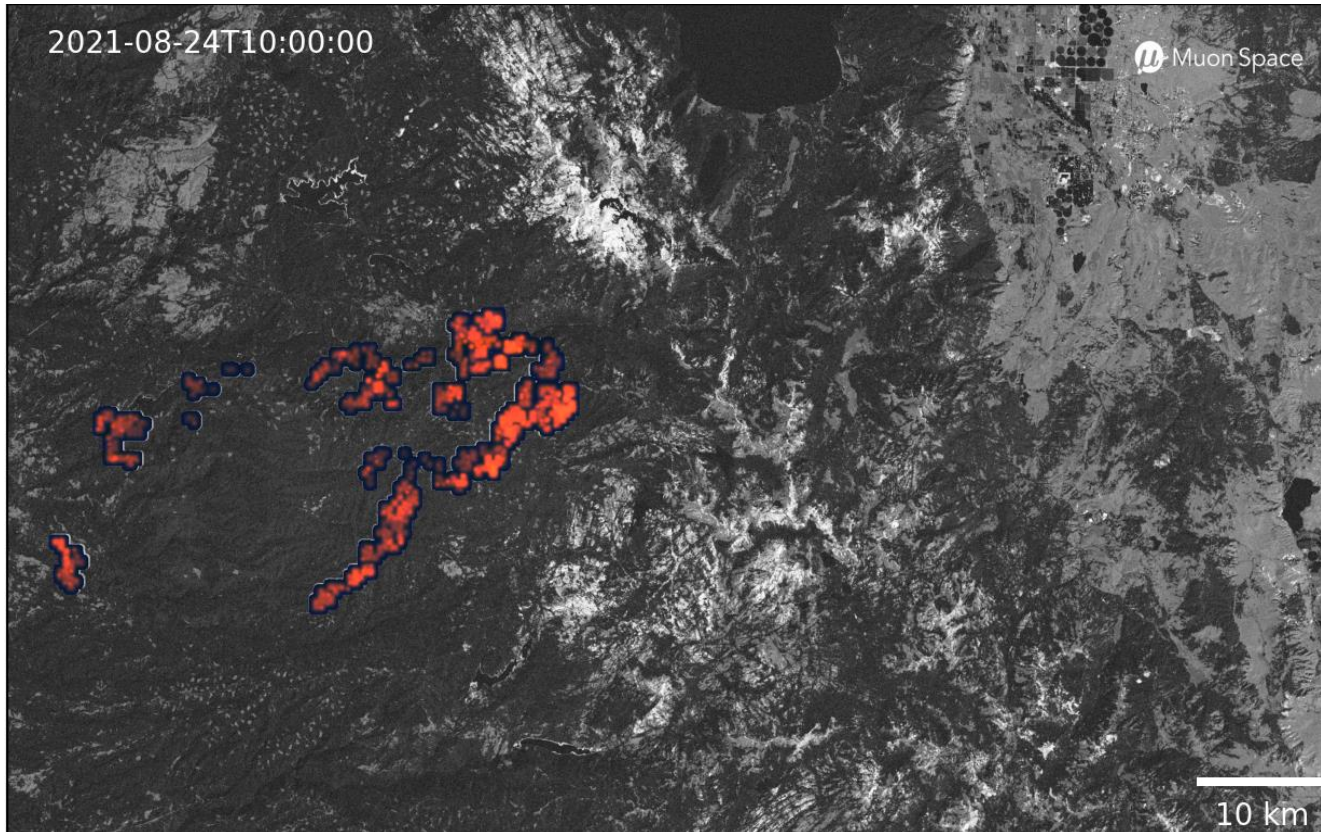
“The synthetic data we generated with Rendered.ai & DIRSIG have been incredibly valuable to a wide range of pre-launch activities on our Tanager program.

We have been able to demonstrate the value we expect to deliver to our hyperspectral customer base ahead of launch.”

-Mark Keremedjiev, Mission Director



Synthetic Data in Action: Muon Space



USE CASE: Developing analytics pre-launch for FireSat, an EO/IR mission to revolutionize wildfire detection & monitoring that empowers first responders to protect communities & the environment.

"The ability to product realistic synthetic MSI observations for our new mission was crucial for our team to build analytics pipelines & engage mission stakeholders before the first satellite was launched."

-Kyle Story, Sr. Manager, IR Data Product





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- Or Visit Booth #2011 for a Live Demo Now



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